Chapter 1 Introduction

1.1. Existing System

The existing system refers to the currently available social media platforms that allow users to share images, interact with posts, and connect with other users. Popular applications such as Instagram provide features like user registration, image posting, likes, comments, and profile management.

In the existing systems, users can upload photos, apply basic filters, like posts, and follow other users. These platforms mainly focus on visual content sharing and social interaction. However, many existing solutions are complex, resource-heavy, and not customizable for learning or small-scale deployment purposes.

Additionally, existing systems often have limitations such as restricted control over data, Hidden internal processes, and limited customization for developers or institutions. For academic or beginner-level projects, these platforms cannot be modified or studied internally because their backend logic and database structure are not accessible.

Therefore, while existing systems are feature-rich and widely used, they are not suitable for understanding internal workflows such as database management, like mechanisms, user authentication, and post-handling logic, which creates a need for a simplified and customizable alternative like PixelAura+.

1.2 Need for the New System

Although existing social media platforms such as Instagram provide advanced features for image sharing and user interaction, they are not suitable for academic learning, experimentation, or small-scale application development. These platforms are closed systems where developers and learners do not have access to internal logic, database structures, or core functionalities.

There is a strong need for a new system that is simple, customizable, and easy to understand while still offering essential social media features. Students and developers require a platform where they can learn how image-based applications work internally, including user authentication, post management, like functionality, and database relationships.

The new system, PixelAura+, is designed to overcome these limitations by providing a lightweight, transparent, and developer-friendly environment. It allows full control over the source code and database, enabling easy modification and enhancement of features based on requirements.

Additionally, PixelAura+ focuses on improved performance, better data control, and a simplified user experience. The system is ideal for learning purposes, project development, and demonstrating core concepts of modern web applications.

Thus, the need for a new system arises to provide an accessible, flexible, and educational alternative to existing large-scale social media platforms

1.3 Objective of the New System

The main objective of the new system, PixelAura+, is to develop an image-based social media application similar to Instagram that enables users to like content and interact with posts in a simple and efficient manner. The system aims to provide essential social networking features while maintaining a clean and user-friendly interface.

The key objectives of PixelAura+ are as follows:

- To allow users to register and securely as per my core project log in to the system.
- To enable users to upload images and create posts.
- To provide like functionality for user interaction with posts.
- To allow users to delete their own posts when required.
- To display real-time like counts for each post.
- To design a responsive and visually appealing user interface.
- To maintain a structured and secure database for users, posts, and likes.
- To create a customizable platform suitable for learning and development purposes.

Overall, PixelAura+ aims to demonstrate the core working principles of a modern social media platform while offering flexibility, simplicity, and scalability for future enhancements.

1.4. Problem Definition

In the current digital era, image-based social media platforms have become an important medium for communication and content sharing. However, existing platforms such as Instagram are complex, closed-source, and not suitable for academic learning or small-scale development. Users and developers do not have access to internal system logic, database structures, or core functional workflows.

For students and beginner developers, understanding how features like image uploading, post management, like mechanisms, and user authentication work at the backend level is difficult due to the lack of transparency in existing systems. Additionally, building a fully functional social media application from scratch requires proper planning, structured design, and efficient data handling, which is often challenging without a simplified reference system.

The problem is to design and develop a customizable, easy-to-understand, and efficient image-like platform that provides essential social media functionalities while allowing full access to the system architecture and database. The system must be secure, user-friendly, and scalable, enabling users to interact with content effectively.

PixelAura+ addresses this problem by offering a lightweight Instagram-like platform that demonstrates core social media functionalities in a controlled and educational environment. Similar to Instagram that enables users to share content and interact with posts in a simple and efficient manner. The system aims to provide essential social networking features while maintaining a clean and user-friendly interface.

The key objectives of PixelAura+ are as follows:

- To allow users to register and securely log in to the system.
- To enable users to upload images and create posts.
- To provide like functionality for user interaction with posts.
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1.5. Project Profile

Project Title	Pixel Aura +
Software Used	VISUAL STUDIO CODE
Browser	CHROME
Front End	HTML, CSS, JavaScript
Back End	MySQL Workbench, Python + Django
Project Description	Social Media Platform
Scope	This project will automate students, small users. Same Like a social Media. Add Post, like, follow/unfollow/Account Privacy. Best for beginner who want know how Instagram works.

1.6. Core Components

The Pixel Aura+ system is composed of several core components that work together to provide a smooth and interactive image-based social media experience. Each component plays a vital role in ensuring system functionality, performance, and usability.

1. User Authentication Module: This component manages user registration and login processes. It ensures secure access to the system by validating user credentials and maintaining session management.

2. User Profile Module: Allows users to view and manage their profile information. It displays user details and their uploaded posts.

3. Post Management Module: This module enables users to upload images, create posts, view posts, and delete their own posts. It handles image storage and post-related data in the database.

4. Like System: The like system allows users to like or unlike posts. It updates the like count dynamically and ensures that a user can like a post only once.

5. Database Management System: Stores and manages all system data, including user details, posts, and likes. It ensures data consistency and efficient retrieval.

6. User Interface (UI): Provides a responsive and visually appealing interface for interacting with the system. It ensures ease of use and smooth navigation.

7. Backend Logic: Handles server-side operations such as data processing, validations, and communication between the frontend and database.

1.7 Assumptions and Constraints

During the design and development of the Pixel Aura+ system, certain assumptions and constraints have been considered to ensure smooth implementation and realistic system expectations.

Assumptions

- Users have basic knowledge of using web-based applications.
- The system will be accessed through a modern web browser.
- Users will have a stable internet connection.
- Each user will have a unique account for accessing the system.
- The system is intended for academic and learning purposes.
- The number of users is limited and manageable.

Constraints

- The system supports only image-based posts.
- Advanced features such as reels, stories, and live streaming are not included.
- Real-time notifications are not implemented.
- The system depends on server availability for proper functioning.
- Security features are limited to basic authentication mechanisms.
- The application is designed for small-scale usage and not for large commercial deployment.

1.8 Advantages and Limitations of the Proposed System

The proposed system, Pixel Aura+, offers several benefits while also having certain limitations due to its scope and design. These are described below:

Advantages

- Simple and user-friendly interface.
- Allows users to upload images and interact with posts through likes.
- Secure user authentication and session management.
- Structured database design for users, posts, and likes.
- Full access to source code and database for learning and customization.
- Lightweight system with faster performance.
- Easy to modify and extend with additional features.
- Suitable for academic projects and skill development.

Limitations

- Limited features compared to commercial social media platforms.
- No support for stories.
- Absence of real-time notifications.
- Basic security mechanisms only.
- Not optimized for large-scale or high-traffic environments.
- No advanced content moderation or admin dashboard.

Chapter 2: Requirement Determination & Analysis

2.1 Requirement Determination

Requirement determination involves identifying the functional and non-functional needs of the system to ensure it meets user expectations. For the Pixel Aura+ project, requirements were identified by analyzing existing image-based social media platforms and understanding the core features required for an Instagram-like application. The focus was on simplicity, usability, and learning-oriented design.

- Functional Requirements

  The functional requirements define the services and operations that the system must perform.

- User Registration and Login

The system must allow users to create an account and log in using valid credentials. This ensures secure access and personalized user interaction within the platform.

- Image Upload and Post Creation

Users should be able to upload images and create posts. Each post must be stored securely in the database along with user details.

- Post Feed Display

The system should display all posts in a structured feed format. Users can view images uploaded by themselves and others.

- Like and Unlike Functionality

Users must be able to like or unlike posts. The system should update and display the like count accurately for each post.

- Post Deletion

Users should have the option to delete their own posts. This ensures content control and better user experience.

- Data Storage and Management

The system should store user, post, and like information in a structured database. This helps maintain data consistency and quick retrieval.

- Non-Functional Requirements

  These requirements describe the qualities and constraints of the system.

- Usability

The application should be simple and easy to use, even for first-time users. Clear navigation and responsive design are required.

- Performance

The system should respond quickly to user actions such as liking or uploading posts. Delays should be minimal to ensure smooth interaction.

- Security

Basic security mechanisms must be implemented to protect user credentials and data. Unauthorized access should be prevented.

- Compatibility

The system should work properly on modern web browsers and devices. This ensures accessibility across different platforms.

2.2 Targeted Users

The Pixel Aura+ system is designed for a specific group of users who require a simple and interactive image-sharing platform for learning and experimentation. The target users are identified based on usability, purpose, and system scope.

- Students

Pixel Aura+ is primarily intended for students who are learning web development and database concepts. It helps them understand how social media platforms work internally.

- Beginner Developers

The system is suitable for beginner-level developers who want to practice building Instagram-like applications. It allows them to explore features such as authentication, image upload, and likes.

- Academic Institutions

Educational institutions can use Pixel Aura+ as a demonstration project. It serves as a practical example for teaching full-stack development concepts.

- Small User Groups

The system can be used by a limited number of users for testing and learning purposes. It is not intended for large-scale commercial use.

- UI/UX Designers

Designers can use the platform to test interface layouts, user experience flows, and visual design concepts for social media applications.

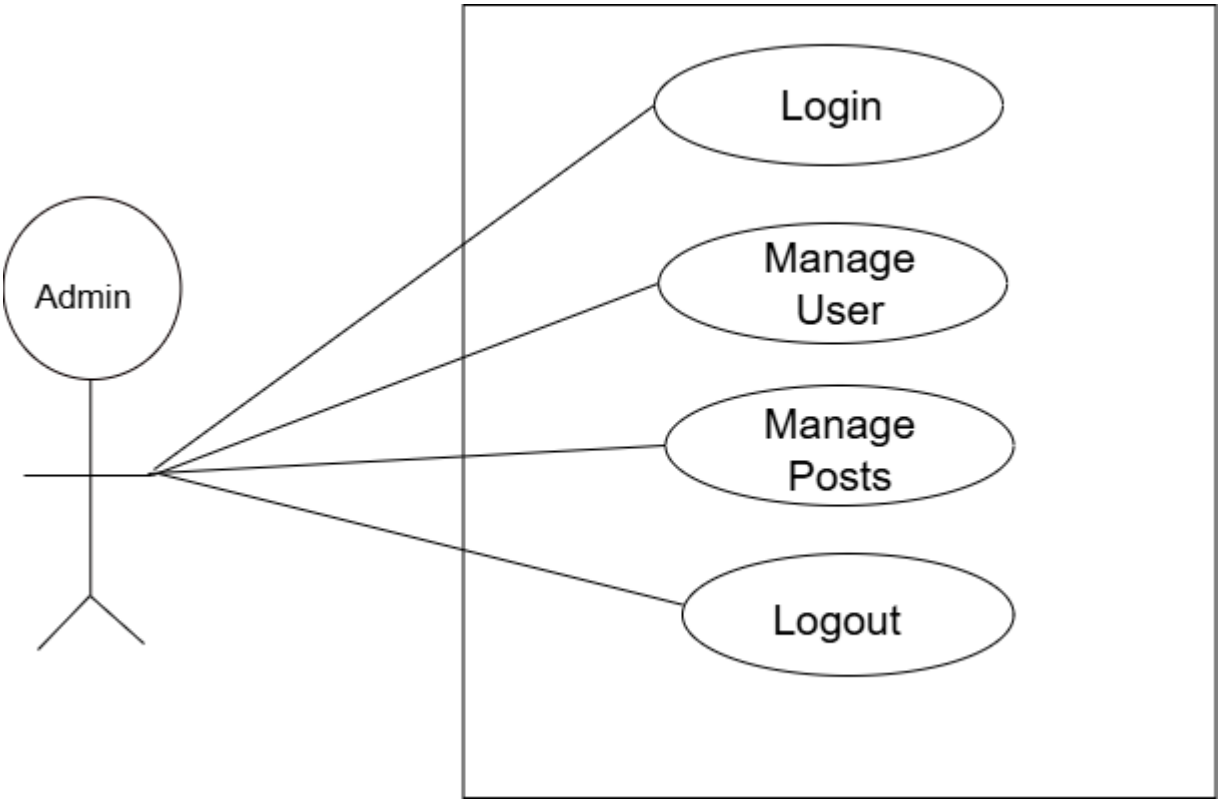
- Software Trainers and Mentors

Trainers and mentors can use Pixel Aura+ to demonstrate real-world system architecture, explain workflows, and guide students in building similar platforms.

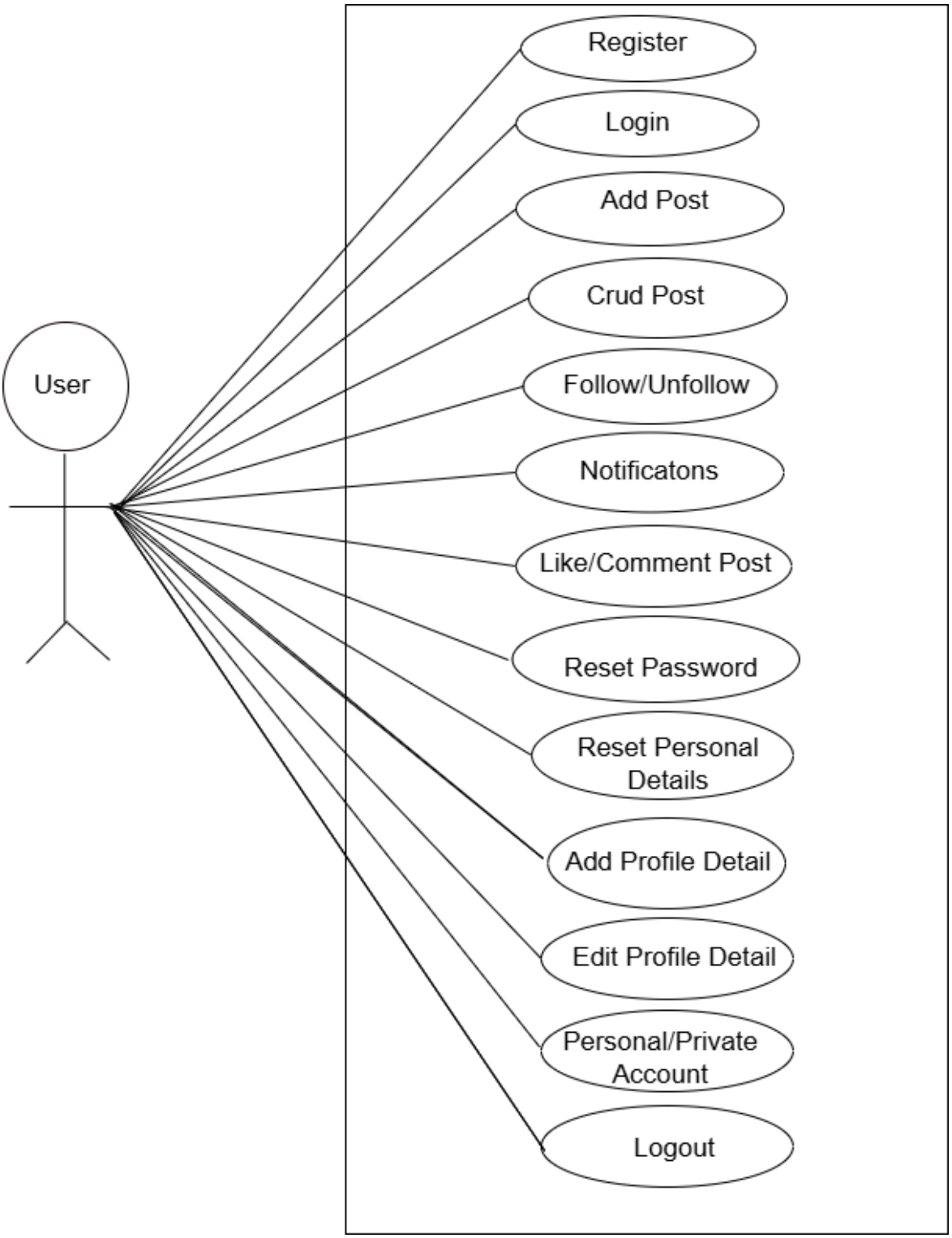
Chapter 3 System Design

3.1 Use Case Diagram

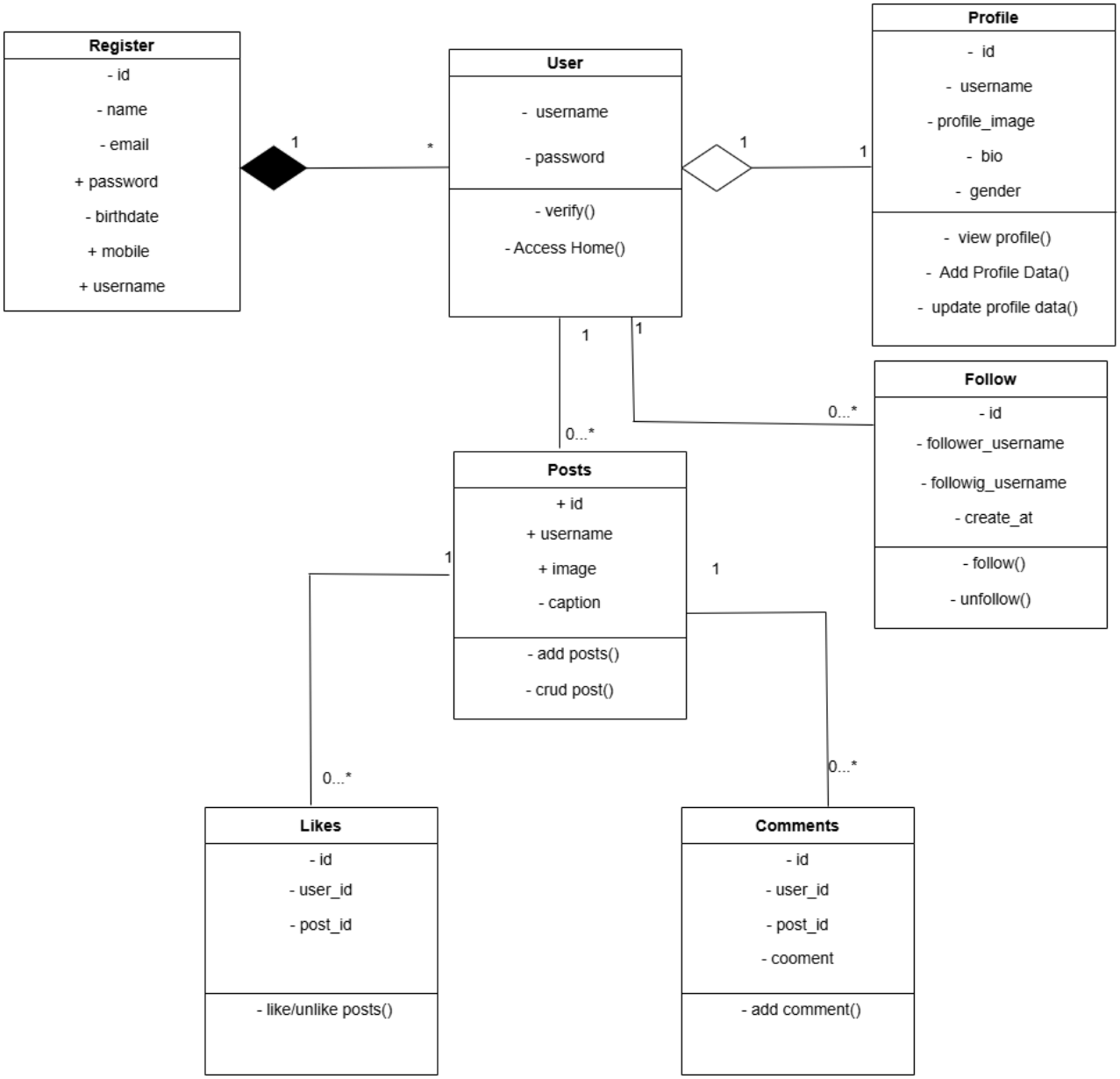
Admin



User/Client

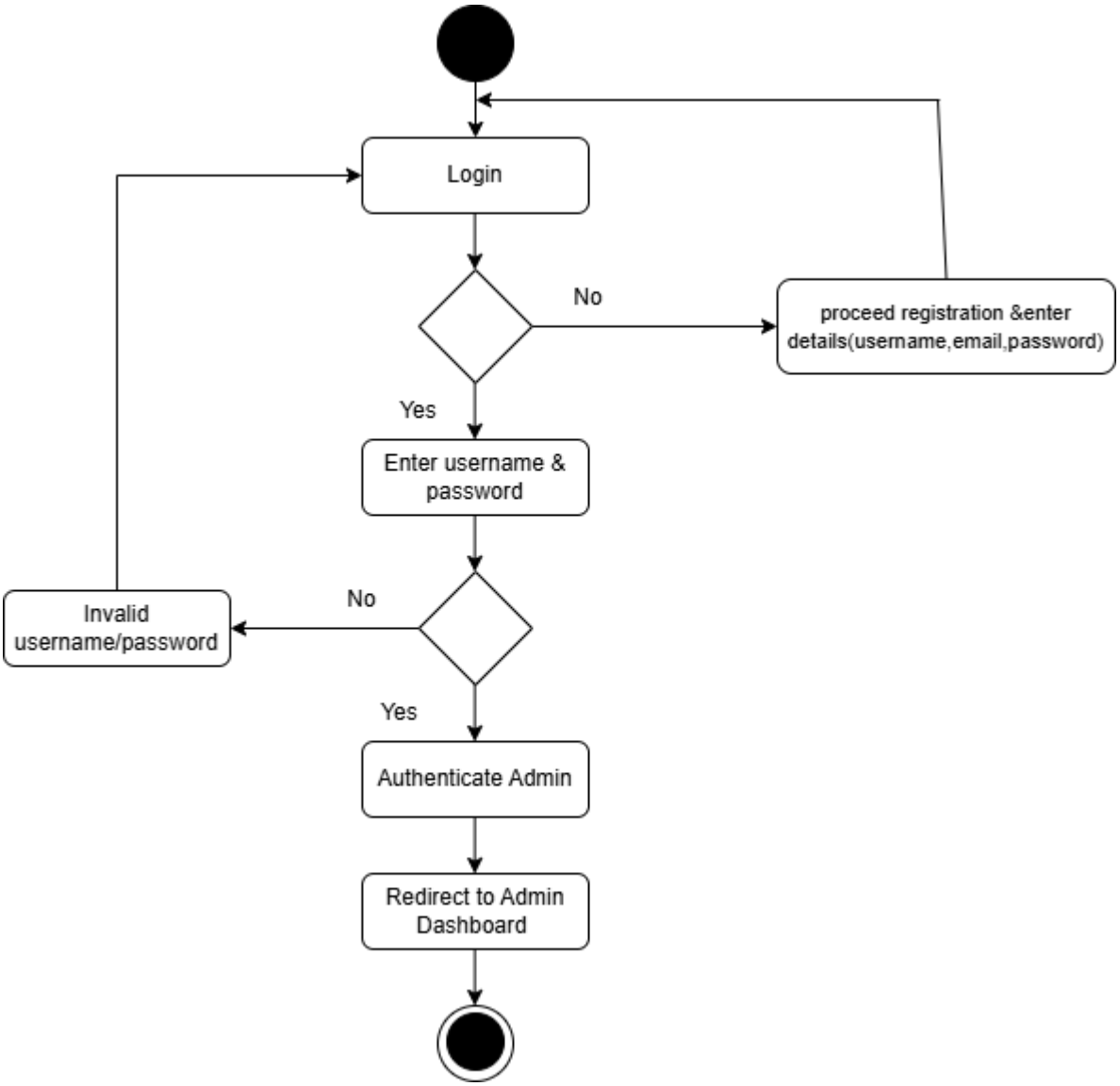

User
Register
Login
Add Post
Crud Post
Follow/Unfollow
Notificatons
Like/Comment Post
Reset Password
Reset Personal Details
Add Profile Detail
Edit Profile Detail
Personal/Private Account
Logout

3.2 Class Diagram

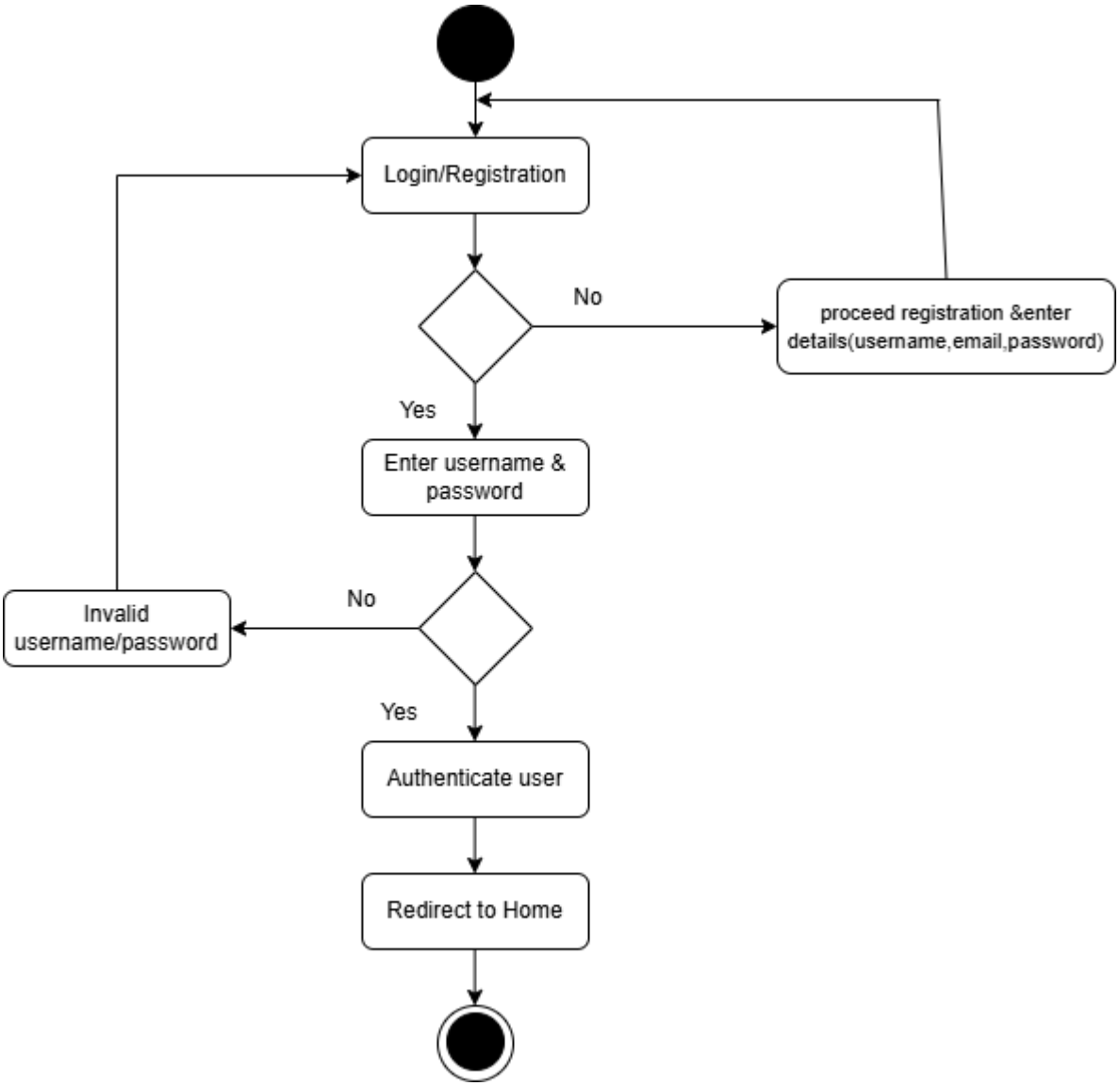


3.3 Activity Diagram

Admin



USER


Login/Registration
No
proceed registration &enter details(username,email,password)
Yes
Enter username & password
No
Invalid username/password
Yes
Authenticate user
Redirect to Home

3.4 Data Dictionary

Client/User

Register

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
name	varchar(45)	No		
email	varchar(45)	No		
password	varchar(40)	No		
birthdate	varchar(45)	No		
mobile	varchar(45)	No		
username	varchar(45)	No		
Is private	Boolean	NO	1	

Posts

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
image	long text	No		
username	varchar(45)	No		
caption	Long text	No		

Profile

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
username	varchar(45)	No		
image	Long text	No		
bio	Long text	No		
gender	varchar(10)	No		

Follows

Column	Type	Null	Default	Comments
id (Primary)	int(11)	No		
Follower_username	varchar(45)	No		
Following_username	varchar(45)	No		
Created_at	timestamp	Yes	current_timestamp()	
Status	varchar(45)	No	accepted	

Likes

Column	Type	Null	Default	Comments
id (Primary)	Int(11)	No		
Post_id	Int(11)	No		
User_id	Int(11)	No		
Created_at	timestamp	Yes	current_timestamp()	

Notifications

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
User_id	Int(11)	No		
message	varchar(255)	No		
Is_read	Tinyint(1)	No		
Created_at	timestamp	Yes	current_timestamp()	

Comments

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
Post_id	Int(10)	No	NULL	
User_id	Int(10)	No	NULL	
Comments	Long text	No	NULL	
Created_at	timestamp	Yes	current_timestamp()	

Chats

Column	Type	Null	Default	Comments
id *(Primary)*	int(11)	No		
sender	varchar(30)	No	NULL	
receiver	varchar(30)	No	NULL	
message	Long text	No	NULL	
Created_at	timestamp	Yes	current_timestamp()	

Chapter 4 Development

4.1 Coding Standards

(User)

Register

```
<!-- logo + brand -->
<img src="{% static 'images/logo.png' %}" class="logo" alt="PixelAura+ logo">
<div class="brand-name">PixelAura+</div>

<!-- SINGLE FORM (2 steps) -->
<form class="register-form" method="post" enctype="multipart/form-data">
    <div class="form-inner">
        {% csrf_token %}
        <input type="hidden" name="step" id="stepField" value="1">

        <!-- ========= STEP 1 ========= -->
        <div class="step step-1 active">
            <h3 class="section-title">Create your account</h3>
            <input type="text" name="name" placeholder="Full name" required>
            <input type="email" name="email" placeholder="Email" required>
            <input type="text" name="mobile" placeholder="Mobile" required>

            <input type="text" name="password" placeholder="Password" required>
            <input type="text" name="confirm_password" placeholder="Confirm password" required>

            <button type="button" class="primary-btn" id="nextBtn">Next</button>
        </div>


        <!-- ========= STEP 2 ========= -->
        <div class="step step-2">
            <h3 class="section-title">More details</h3>

            <div class="extra-grid">
```

Login

```
<form class="login-form" method="post">
    {% csrf_token %}

    <input type="text" name="username" placeholder="Username" required>
    <input type="text" name="password" placeholder="Password" required>

    <button type="submit" class="login-btn">Log in</button>
</form>
</div>


<div class="bottom-card">
    <span>Don't have an account?</span>
    <a href="{% url 'register' %}">Register</a>
</div>

</div>


<div class="right-side">
    <div class="phone-stack">
        <div class="phone-img img1">
            <img src="{% static 'images/img1.jpeg' %}" alt="preview 1">
        </div>
        <div class="phone-img img2">
            <img src="{% static 'images/img2.jpg' %}" alt="preview 2">
```

Home

```
<div class="suggested-panel">
  <div class="suggest-header">
    <span>Suggested for you</span>
  </div>

  <div class="story-suggest-wrapper">
    <!-- ✅ LEFT SIDE: HORIZONTAL SUGGESTED CREATORS -->
    <div class="story-logos-row">
      {% for u in ulist %}
      <div class="logo-wrapper">
        <a href="{% url 'suggested_user_profile' u.username   %}">
          {% if u.image %}
          <img src="{{ u.image }}" class="suggest-img" />
          {% else %}
          <img
            src="{% static 'images/default-user.png' %}"
            class="suggest-img"
          />
          {% endif %}
        </a>
        <p class="logo-name">{{ u.username }}</p>
      </div>
      {% endfor %}
    </div>

    <!-- ✅ RIGHT SIDE: MAIN USER -->
    <div class="main-user-box">
      <a href="{% url 'profile' %}" class="profile-link">
        {% if profile.image %}
        <img
          src="{{ profile.image }}"
          class="logo-wrapper user-logo suggest-img"
```

Settings

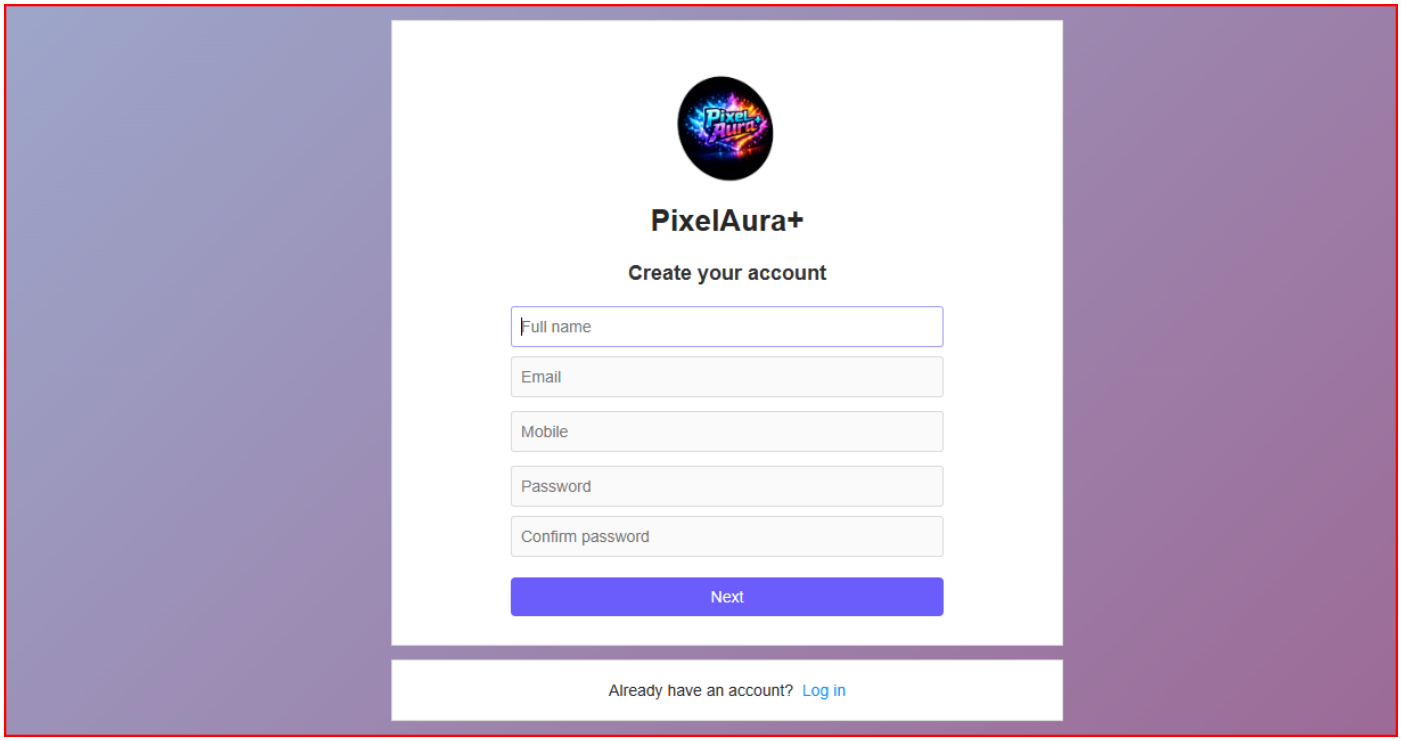
```
ra > PixelAuraplus > templates > setting.html
<div class="settings-panel">
  <div class="accounts-card">
    <div class="meta-row">
    </div>

    <h4>Accounts Center</h4>
    <p class="muted">
      Manage your connected experiences and account settings across LensORa technologies.
    </p>

    <ul class="accounts-links">
      <li>
        <a href="{% url 'personal_detail_s' %}" class="account-link">
          <i class='bx bx-user'></i>
          Personal details
        </a>
      </li>
    </ul>


  </div>

  <!-- ✅ SECTION 1 -->
  <p class="section-title">How you use Instagram</p>
  <ul class="settings-list" id="editProfileTab">
    <li class="active">
      <a href="javascript:void(0)">
        <i class='bx bx-edit'></i>
        Edit profile
      </a>
    </li>

  </ul>

  <!-- ✅ SECTION 2 -->
  <p class="section-title">Who can see your content</p>
  <ul class="settings-list" id="privacyTab">
    <li>
      <a href="javascript:void(0)">
```

Profile

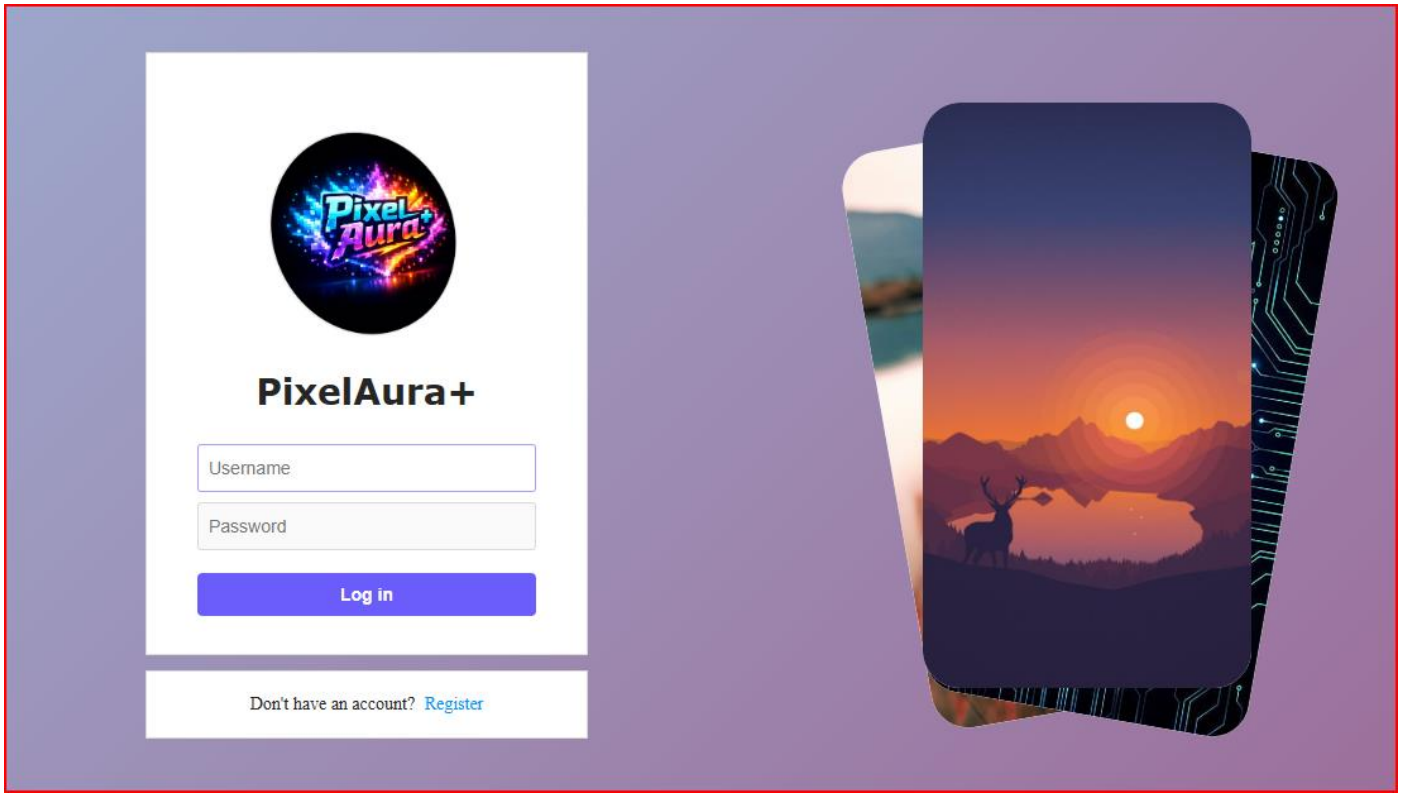
```
<body>
  <div class="profile-container">
    <div class="profile-sticky">
      <div class="user-row">
        <!-- ✅ PROFILE IMAGE WITH CAMERA & DEFAULT ICON -->
        <div class="profile-pic-wrapper">

          {% if profile.image %}
          <img src="{{ profile.image }}" class="profile-img">
          {% else %}

          <img src="{% static 'images/default-user.png' %}" class="profile-img">
          {% endif %}
        </div>


        <!-- ✅ USERNAME + NAME + SETTINGS ICON -->
        <div class="user-right">

          <div class="user-top-row">
            <div class="user-text">
              {% for login in userlogin %}
              <h2 class="username">{{ login.username }}</h2>
              <span class="full-name">{{ login.name }}</span>
              {% endfor %}
            </div>

          </div>
```

4.2 Screenshots

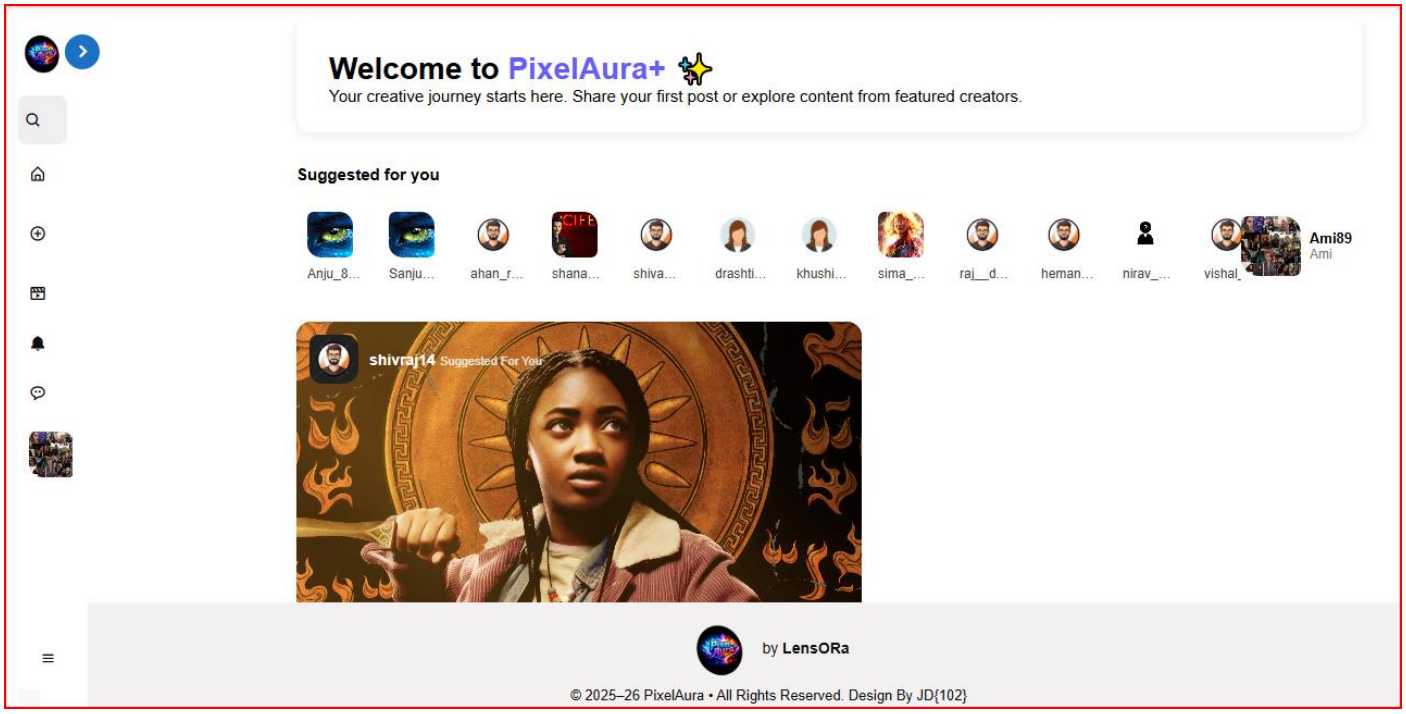
Register



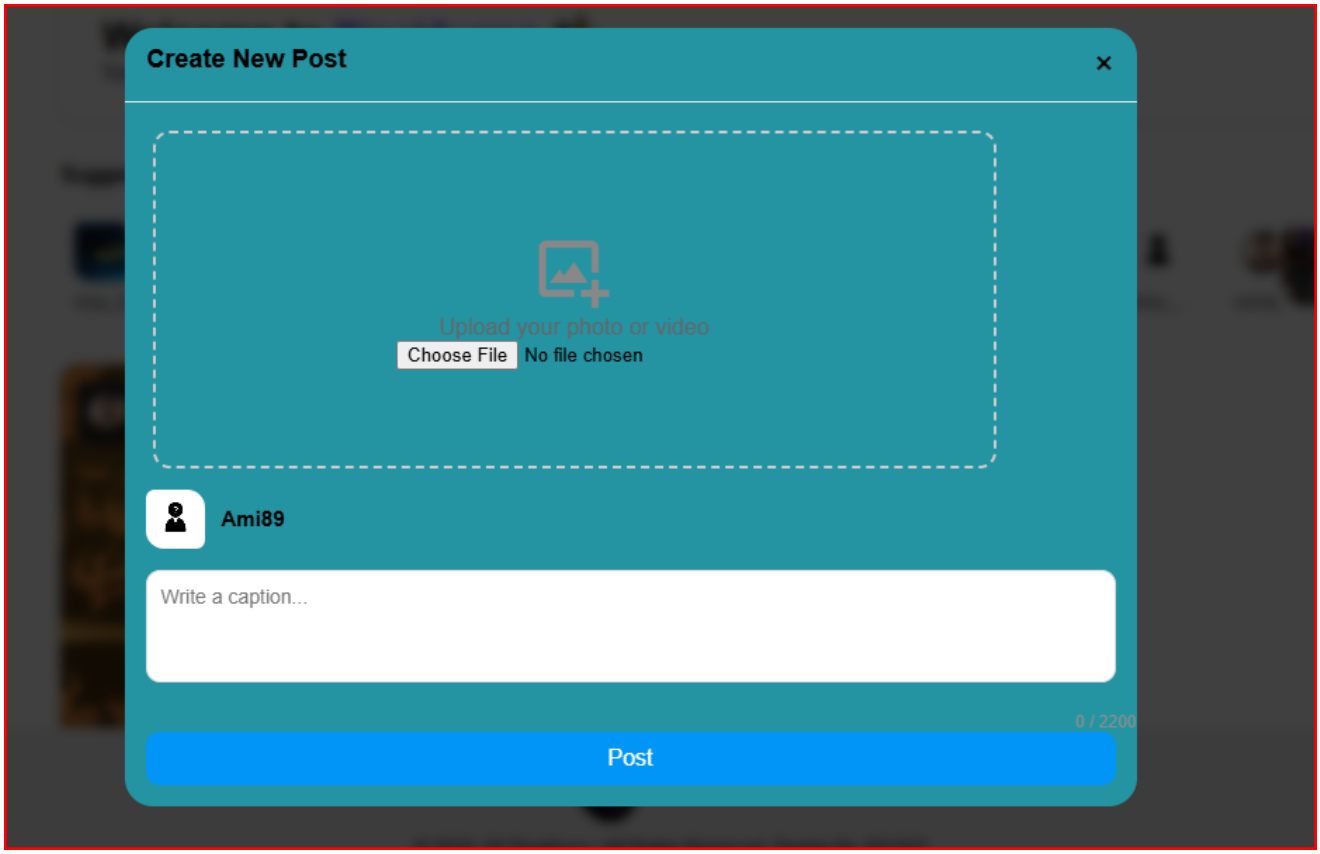
Login


PixelAura+
Username
Password
Log in
Don't have an account? Register

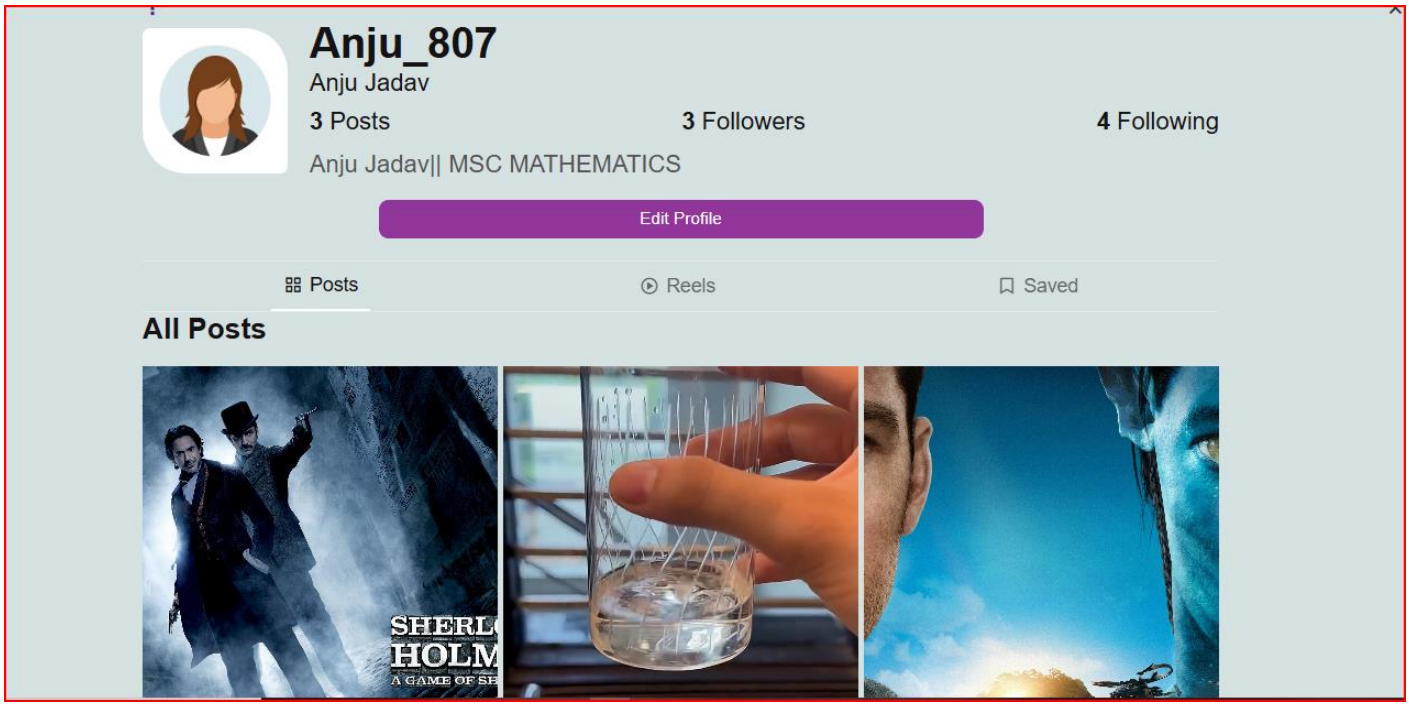
Home


Welcome to PixelAura+
Your creative journey starts here. Share your first post or explore content from featured creators.
Suggested for you
Anju_8...
Sanju...
ahan_r...
shana...
shiva...
drashti...
khushi...
sima_...
raj__d...
heman...
nirav_...
vishal_
Ami89
Ami
shivraj14 Suggested For You
by LensORa
© 2025–26 PixelAura • All Rights Reserved. Design By JD{102}

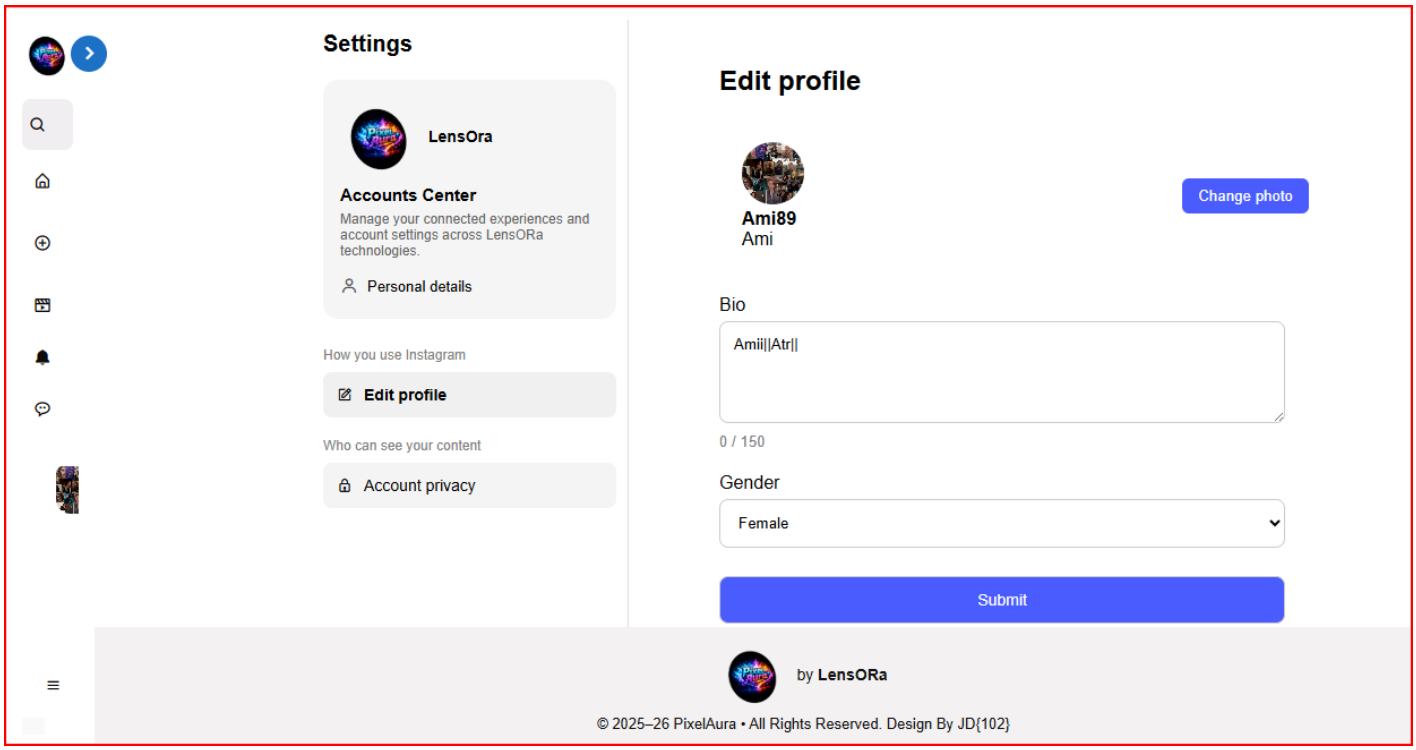
Add Post


Create New Post
×
Upload your photo or video
Choose File
No file chosen
Ami89
Write a caption...
0 / 2200
Post

Profile


Anju_807
Anju Jadav
3 Posts
3 Followers
4 Following
Anju Jadav|| MSC MATHEMATICS
Edit Profile
Posts
Reels
Saved
All Posts

Settings


Settings
LensOra
Accounts Center
Manage your connected experiences and account settings across LensORa technologies.
Personal details
How you use Instagram
Edit profile
Who can see your content
Account privacy
Edit profile
Ami89
Ami
Change photo
Bio
Amii||Atr||
0 / 150
Gender
Female
Submit
by LensORa
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Chapter 5: Agile Documentation

Agile methodology is an iterative and incremental approach to software development that focuses on flexibility, collaboration, and continuous delivery of valuable features. The Hostel Management System project has been developed using agile principles to ensure effective communication, quick adaptation to changes, and frequent delivery of functional modules.

5.1 Agile Project Charter

Project Title: PixelAura+

Project Description:
The PixelAura+ PixelAura+ is a web-based social media application developed using Python with Django and MySQL Workbench, designed to enable users to create profiles, like content, comment, and follow other users. The system aims to simplify social interaction, enhance user engagement, and provide a secure platform for digital content sharing. It reduces manual effort in managing user interactions and ensures efficient handling of user data and platform activities.

Project Objectives:

- Automate user account management and profile handling.
- Simplify content sharing, post creation, and interaction processes.
- Provide a transparent system for likes, comments, and follower management.
- Enhance accessibility through a secure and user-friendly online platform. Accessibility through a secure online platform.

Stakeholders:

- **Admin:** Manages User, Content.
- **User/Client:** Add Post, Like and Comment, Follow, Add profile, chat

Scope:
The PixelAura+ project aims to provide a social media platform that allows students and small users to share posts, interact with others, and manage profiles in a secure and user-friendly environment.
Team Members:
Product Owner: [Pro. Sunny Goswami]
Scrum Master: [Ajay Kathi]

Developers: [Ajay Kathi]
QA/Testers: [Ajay Kathi]
Assumptions: Project will follow Scrum, with 2-week sprints.
Constraints: Limited budget, must be completed in 4 months.

5.2 Agile Roadmap / Schedule

Iteration / Sprint	Duration	Main Deliverables
Sprint 1	Week 1	Requirement gathering, database design, and setup.
Sprint 2	Week 2	UI/UX design, database schema design.
Sprint 3	Week 3	User registration, login, profile module
Sprint 4	Week 4	Post creation, feed display, image upload
Sprint 5	Week 5	Likes, comments, follow system
Sprint 6	Week 6	Reel, Suggests User
Sprint 7	Week 7	Profile, Chat, Notification
Sprint 8	Week 8	Admin

5.3 Agile Project Plan

Purpose:
This Agile Project Plan outlines how the PixelAura+ social media platform will be developed using Agile Scrum practices, ensuring iterative progress, continuous feedback, and timely delivery of system features.

Include:

- Methodology: Scrum
- Sprint Duration: 1 weeks
- Total Sprints: 8
- Tools:
    - Jira — Task tracking and sprint management
    - Trello — Workflow organization
    - GitHub — Version control and collaboration
    - Zoom — Team communication and sprint meetings

- Key Meetings:
    - Sprint Planning - Define sprint goals and assign tasks
    - Daily Standups - Track daily progress and resolve blockers
    - Sprint Reviews & Retrospectives- Evaluate sprint outcomes and improve workflow

- Roles and Responsibilities:
    - Scrum Master: Facilitates Agile processes, removes obstacles, and ensures Scrum practices are followed
    - Product Owner: Manages product backlog, prioritizes features, and gathers user requirements
    - Developers: Design, build, and integrate system functionalities
      QA (Quality Assurance): Tests features, identifies bugs, and ensures system quality

5.4 Agile User Story (Minimum 3 Tasks)

User Story 1: User/Client Registration and Login

As a User/Client, I want to register and log in so that I can access my home Page.
Tasks:

1. Create registration form for User.
2. Implement login validation and session management.
3. Display Home Page after successful login.

User Story 2: Add Post

As a User/Client, I want to Add Post. Which is save to Database.
Tasks:

1. Create Add Post Page.
2. User can add Post, Write Caption.

User Story 3: Setting

As a User, I want to Edit Bio, Change My Personal Detail, Change Password, and Private Account.

Tasks:

1. Create a setting Link or Page.
2. Where Links for Profile, Personal Detail, Account Privacy.
3. User Do Edit Bio, Change My Personal Detail.

User Story 5: Like, Comment, Follow System

As a user I want to like, comment and follow other user.

Tasks:

1. Implement like and comment features for posts.
2. Develop a follow/unfollow system between users.
3. Display interaction counts (likes, comments, followers).

User Story 4: Admin

As an Admin, I want To Mange User, Posts and Login to My Dashboard.
Tasks:

1. As Per Technology Stack Login Page is Default.
2. Authenticate admin and redirect to admin Dashboard.
3. User Admin Functionality.

5.5 Agile Release Plan

Release	Sprint	Features Included
Release 1	Sprint 1–2.	Requirement gathering, project setup, UI/UX design, database schema finalization.
Release 2	Sprint 3–4	User registration, login, authentication, profile setup, post creation, feed display.
Release 3	Sprint 5–6	Likes system, comments module, follow/unfollow system, reels feature, user suggestions.
Release 4	Sprint 7–8	Chat/messaging, notifications, admin panel, testing, bug fixing, deployment, final documentation.

5.6 Agile Sprint Backlog

Purpose:

The Sprint Backlog lists the tasks and user stories that must be completed during the current sprint. It helps the team track progress, assign responsibilities, and ensure timely delivery of sprint goals.

Release	Task	Assignee	Status
Post Creation	Design post upload UI.	Dev A	In Progress
Post Creation	Implement image upload feature.	Dev B	To Do
Feed Display	Fetch posts from database	Dev C	In Progress
Feed Display	Display posts in user feed	Dev A	To Do
User Interaction	Add like and comment buttons	Dev B	To Do

5.7 Agile Test Plan

Purpose:
The Agile Test Plan defines how testing will be conducted in each sprint to ensure the PixelAura+ system is reliable, secure, and user-friendly. Testing is performed continuously to detect defects early and maintain high software quality.

Test Types:

- Unit Testing — Test individual components and functions
- Integration Testing — Verify interaction between modules
- UI Testing — Ensure proper layout, responsiveness, and usability
- User Acceptance Testing (UAT) — Confirm system meets user expectations.

Testing Tools:

- Manual Test Cases — Functional and usability testing

Sprint-wise Testing Plan:

- Sprint 1: Test user registration and login functionality
- Sprint 2: Test profile creation and account management
- Sprint 3: Test post creation, image upload, and feed display
- Sprint 4: Test likes, comments, and follow system
- Sprint 5: Test reels, search, and user suggestion features
- Sprint 6: Test chat, notifications, and messaging system
- Sprint 7: Perform system integration testing
- Sprint 8: Conduct final UAT, performance testing, and bug fixing

Chapter 6 Proposed Enhancements

1. Automated Room Allocation System

PixelAura+ successfully implements core social media functionalities such as user registration, profile management, post sharing, likes, comments, and following, several enhancements can be introduced in the future to improve performance, usability, security, and scalability. These proposed upgrades aim to make the platform more interactive, intelligent, and competitive with modern social networking applications.

2. Advanced Messaging and Real-Time Chat

A future enhancement includes integrating a real-time messaging system that allows users to send instant messages, share media, and create group chats.

This feature can be implemented using Firebase, enabling fast communication, typing indicators, message read receipts, and online status visibility, which significantly improves user engagement. Digital Complaint & Maintenance Portal

3. Stories and Live Streaming Feature

A Stories module can be added to allow users to share temporary photos and videos that disappear after 24 hours.

Additionally, a Live Streaming feature can allow creators to broadcast live content, interact with followers in real time, and increase platform activity and audience interaction.

4. AI-Based Content Recommendations

Artificial Intelligence (AI) can be integrated to provide personalized content recommendations based on user interests, interaction history, and behavior.

This can help users discover relevant posts, trending topics, suggested profiles,

and hashtags, improving content visibility and platform engagement.

5. Enhanced Privacy and Security Controls

To strengthen **user trust and data protection**, additional privacy options can be introduced, including:

- Private and public account settings
- Post visibility control (friends-only, followers-only)
- Two-factor authentication (2FA)
- Account activity logs
- Advanced reporting and blocking tools

6. Push Notifications and Alerts

A push notification system can be added to alert users about:

- New followers
- Likes and comments
- Mentions and tags
- Direct messages
- Platform announcements

7. Content Analytics and Creator Dashboard

A Creator Dashboard can provide insights into:

- Post reach and impressions
- Engagement metrics (likes, comments, shares)
- Follower growth trends
- Best-performing content

This feature helps content creators analyze performance and improve content strategies.

8. Monetization and Advertisement System

PixelAura+ can introduce monetization options, such as:

- Sponsored posts
- In-app advertisements
- Paid subscriptions
- Creator tipping systems

These features can support **platform revenue generation** and creator income opportunities.

Chapter 7 Conclusion

The PixelAura+ project successfully demonstrates the design and development of a modern social media platform that enables users to register, create profiles, like content, comment, and follow other users. The system has been built using Python with Django for the backend and MySQL Workbench for database management, ensuring a robust, scalable, and efficient architecture.

Throughout the development process, the project followed agile methodology, allowing features to be built incrementally through multiple sprints. This approach supported continuous improvement, rapid feedback integration, and better alignment with user requirements. Core functionalities such as authentication, post management, interaction systems, and profile handling were implemented effectively, resulting in a user-friendly and interactive platform.

The project also highlights the importance of software engineering best practices, including modular development, database normalization, testing strategies, and version control. These practices contribute to improved code maintainability, system reliability, and future scalability.

In conclusion, PixelAura+ serves as a strong foundation for a full-scale social networking application. With future enhancements such as AI-based recommendations, real-time messaging, mobile app development, and advanced security features, the platform has the potential to evolve into a feature-rich and competitive social media solution. This project not only fulfills its intended objectives but also provides valuable learning and practical experience in web development, database management, and agile project execution.

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